

PHYS 487: Lecture # 21

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Lecture Span

1. More quantum stuff

Density Matrix

Density matrix is defined as

$$\rho = |\psi\rangle\langle\psi| \quad (1)$$

Note that ρ_{ii} are all real values and

$$\sum_i \rho_{ii} = 1 \iff \text{Tr}(\rho) = 1 \quad (2)$$

Moreover,

$$\rho_{ij} = \rho_{ji}^* \quad (3)$$

Imperfect Knowledge

If we want to move the system into a state $|+\rangle$ but there is some variation from the desired state. Then how do we describe this? Use the density matrix

$$\rho = \frac{1}{N} \sum_{\chi} |\psi_{\chi}\rangle\langle\psi_{\chi}| \quad (4)$$

for mixed states. That is $|\psi_{\chi}\rangle \neq |\psi_i\rangle$ for all i, χ . Here, we don't know $|\psi_{\chi}\rangle$ exactly. Then we can represent everything with an average vector. Here,

1. totally mixed: average is 0
2. totally pure: average is 1
3. a bit of both: average $\in (0, 1)$

Pure states have a basis where

$$\rho^2 = \rho \quad (5)$$

This is also called an Idempotent matrix. Also in pure states,

$$\text{Tr}[\rho^2] = \text{Tr}[\rho] = 1 \quad (6)$$

But for mixed states

$$\text{Tr}[\rho^2] < \text{Tr}[\rho] < 1 \quad (7)$$

Then we can define a linear entropy (mixing) as

$$1 - \text{Tr}[\rho^2] \quad (8)$$

We can also define measurements,

$$\text{Pr}[\chi] = |\langle \chi | \psi \rangle|^2 = \langle \chi | \rho | \chi \rangle \quad (9)$$

And an expectation value

$$\langle A \rangle = \text{Tr}[\rho A] \quad (10)$$

And a time evolution on ρ as

$$\rho = -\frac{i}{\hbar} [\rho, H] \quad (11)$$